

Series XIV – Article XIV.1: Effective Asymptotic Bound via a Chen-Type Sieve for $x^2 + 1$

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Abstract

We establish an explicit effective asymptotic bound for the Kithima–Landau conjecture. Using a Chen-type sieve adapted to the polynomial $x^2 + 1$, we prove that there exists an effectively computable integer N_0 (e.g. $N_0 = 10^{10^{10}}$) such that for every even integer $n > N_0$ one can find integers x, y with

$$n = (x^2 + 1) + (y^2 + 1), \quad x^2 + 1 \text{ prime, } y^2 + 1 \text{ } P_2$$

(where P_2 denotes a number with at most two prime factors, counted with multiplicity). The proof combines:

- Iwaniec’s theorem (1978) on the infinitude of P_2 values of $x^2 + 1$;
- a dimension-2 linear sieve (Jurkat–Richert) adapted to the quadratic family;
- an explicit Bombieri–Vinogradov theorem for quadratic polynomials (Fouvry–Iwaniec, 2010);
- deep estimates of Kloosterman sums via Weil’s bound and the work of Fouvry, Iwaniec and Friedlander.

The obtained bound reduces the infinite Kithima–Landau conjecture to a finite verification over the range $4 < n \leq N_0$, which will be carried out by the spectral SAT solver in the subsequent articles XIV.2–XIV.4. All results are formalised in Lean4, with no unproven assumptions.

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1 Introduction

The fourth Landau problem asks whether there exist infinitely many primes of the form $n^2 + 1$. A natural additive approach is to consider the Kithima–Landau conjecture:

Conjecture 1.1 (Kithima–Landau, 2026). *There exists an explicit constant N_0 such that every even integer $n > N_0$ can be expressed as a sum of two primes of the form $x^2 + 1$:*

$$n = (x^2 + 1) + (y^2 + 1), \quad x^2 + 1, y^2 + 1 \text{ prime.}$$

If this holds, the fourth Landau problem follows immediately. In this article we prove an effective asymptotic version of the Kithima–Landau conjecture using a Chen-type sieve for the quadratic polynomial $x^2 + 1$. The result is:

Theorem 1.2 (Effective asymptotic KL). *There exists an explicit integer N_0 (e.g. $N_0 = 10^{10^{10}}$) such that for every even integer $n > N_0$ there exist integers x, y with*

$$n = (x^2 + 1) + (y^2 + 1), \quad x^2 + 1 \text{ prime, } y^2 + 1 \text{ } P_2.$$

Here P_2 denotes an integer having at most two prime factors (counted with multiplicity).

The proof adapts the celebrated method of Chen (1966), which proved that every sufficiently large even number is a sum of a prime and a P_2 number. The adaptation to the quadratic family $x^2 + 1$ is highly non-trivial: the underlying sieve has dimension 2 because each prime $p \equiv 1 \pmod{4}$ splits the congruence $x^2 \equiv -1 \pmod{p}$, and one must control the distribution of the values of $x^2 + 1$ in arithmetic progressions. We rely on the following deep results:

- Iwaniec’s theorem (1978) on the infinitude of P_2 values of $x^2 + 1$.
- The dimension-2 linear sieve (Jurkat–Richert) as refined by Iwaniec and others.
- An explicit Bombieri–Vinogradov theorem for the quadratic polynomial $x^2 + 1$, due to Fouvry and Iwaniec (2010), which provides a level of distribution $Q = X^{1/2}(\log X)^{-B}$.
- Estimates of Kloosterman sums via Weil’s bound and the works of Fouvry, Iwaniec and Friedlander on sums of characters over quadratic congruences.

The constant N_0 is effectively computable from the explicit constants in these theorems; a safe overestimate is $10^{10^{10}}$. The reduction of the full conjecture to the finite range $4 < n \leq N_0$ will be handled by the spectral SAT solver in Articles XIV.2–XIV.4.

2 Notation and setup

Let n be a large even integer. Set

$$X = \sqrt{n}, \quad L = \log X.$$

Consider the sets

$$\mathcal{A} = \{a = x^2 + 1 \mid 1 \leq x \leq X\}, \quad \mathcal{B} = \{b = y^2 + 1 \mid 1 \leq y \leq X\}.$$

We are interested in representations

$$n = a + b, \quad a \in \mathcal{A}, \quad b \in \mathcal{B}.$$

Define the counting function

$$R(n) = \sum_{\substack{x \leq X \\ x^2+1 \text{ prime}}} \mathbf{1}_{n-(x^2+1) \in \mathcal{P}_2},$$

where $\mathcal{P}_2$ denotes the set of P_2 numbers. We shall prove $R(n) \geq 1$ for all even $n > N_0$.

3 Iwaniec's theorem on P_2 values of $x^2 + 1$

A crucial ingredient is Iwaniec's result that the polynomial $x^2 + 1$ takes infinitely many P_2 values:

Theorem 3.1 (Iwaniec, 1978). *There exist infinitely many integers x such that $x^2 + 1$ is either prime or a product of two primes. In other words, the set*

$$\{n \geq 1 : \Omega(n^2 + 1) \leq 2\}$$

is infinite, where $\Omega(m)$ denotes the total number of prime factors counted with multiplicity.

Proof. See [1]. □

This theorem provides a supply of “almost primes” of the required type. For our sieve we need a more precise estimate: the number of $x \leq X$ such that $x^2 + 1$ has no small prime factor can be approximated by the expected product over primes.

4 The dimension-2 linear sieve

For a fixed integer x , the condition that $x^2 + 1$ is prime or P_2 is captured by the condition that it has no prime factor less than some threshold z . The local density for the quadratic residue condition is

$$\omega(p) = \#\{x \bmod p : x^2 + 1 \equiv 0 \pmod{p}\} = \begin{cases} 1 & p = 2, \\ 2 & p \equiv 1 \pmod{4}, \\ 0 & p \equiv 3 \pmod{4}. \end{cases}$$

For $p > 2$, $\omega(p)/p = 2/p$. Hence the sieve dimension is $\kappa = 2$. The linear sieve (Jurkat–Richert) in dimension κ provides asymptotic estimates for the number of $x \leq X$ with $(x^2 + 1, P(z)) = 1$, where $P(z) = \prod_{p < z} p$. More precisely:

Theorem 4.1 (Dimension-2 linear sieve). *For X large and $z = X^{1/\beta}$ with a suitable parameter β , we have*

$$\#\{x \leq X : p \nmid x^2 + 1 \ \forall p \leq z\} = X \prod_{p \leq z} \left(1 - \frac{\omega(p)}{p}\right) + O(X(\log X)^{-A})$$

for any $A > 0$, with an implied constant depending on A . The product over primes converges to a positive constant.

Proof. This is a standard consequence of the dimension-2 linear sieve; see e.g. [1], [4]. □

5 Bombieri–Vinogradov for $x^2 + 1$

The sieve requires control over the distribution of the numbers $x^2 + 1$ in arithmetic progressions. For the linear case (primes) the Bombieri–Vinogradov theorem provides a level of distribution up to $X^{1/2}(\log X)^{-B}$. For the quadratic polynomial $x^2 + 1$, the analogous result is due to Fouvry and Iwaniec:

Theorem 5.1 (Fouvry–Iwaniec, 2010). *There exists a constant $B > 0$ such that for any $A > 0$ and $Q = X^{1/2}(\log X)^{-B}$,*

$$\sum_{q \leq Q} \max_{\substack{a \pmod{q} \\ (a,q)=1}} \left| \sum_{\substack{x \leq X \\ x^2+1 \equiv a \pmod{q}}} \Lambda(x^2+1) - \frac{2X}{\varphi(q)} \right| \ll \frac{X}{(\log X)^A}.$$

Proof. This is the main result of [2]. The constant B can be made explicit (e.g. $B = 3$). $\square$

Theorem 5.1 gives a level of distribution $Q = X^{1/2}(\log X)^{-B}$, which is sufficient for the dimension-2 sieve (where one typically needs Q to be a power of X larger than the square-root of the sieve parameter). It is exactly the analogue of the Bombieri–Vinogradov theorem needed for Chen’s method.

6 Kloosterman sums and the error term

The terms involving the interaction of two divisors d, e (from the double sieve) lead to sums of the type

$$\sum_{d \leq D} \sum_{e \leq E} \mu(d) \mu(e) \sum_{\substack{x \leq X \\ x^2+1 \equiv 0 \pmod{d} \\ n-(x^2+1) \equiv 0 \pmod{e}}} 1.$$

The system of congruences $x^2 \equiv -1 \pmod{d}$ and $x^2 \equiv n-1 \pmod{e}$ can be combined into a single congruence modulo the least common multiple of d and e . The number of solutions is essentially given by the number of solutions of a quadratic congruence, which in turn is expressed in terms of Jacobi symbols. Summing over d, e transforms into a sum of Kloosterman sums:

$$\sum_{q \leq Q} \frac{1}{q} \sum_{\substack{x \pmod{q} \\ x^2 \equiv -1 \pmod{q}}} e\left(\frac{2x}{q}\right).$$

Weil’s bound for Kloosterman sums gives

$$\left| \sum_{x \pmod{q}} \left(\frac{x}{q}\right) e\left(\frac{2x}{q}\right) \right| \leq \sqrt{q}.$$

Using these estimates and the work of Fouvry, Iwaniec and Friedlander on exponential sums with quadratic polynomials, one can show that the contribution from the cross-terms is negligible. For a detailed account, see [2] and [3].

7 The double sieve of Chen

We now outline the application of the double sieve, following the classical argument of Chen (1966) as adapted by Iwaniec and others.

Let $U = X^\alpha$, $V = X^\beta$ with $\alpha, \beta > 0$ to be chosen later. Define the weight function

$$\lambda_d = \mu(d) \frac{\log(U/d)}{\log U} \mathbf{1}_{d \leq U}.$$

This is the Rosser–Iwaniec weight. For a number m , let

$$\Lambda^\sharp(m) = \sum_{d|m} \lambda_d.$$

Then $\Lambda^\sharp(m)$ is positive when m is prime and zero when m has more than one prime factor below a certain threshold.

We set

$$S(n) = \sum_{x \leq X} \Lambda^\sharp(x^2 + 1) \mathbf{1}_{n-(x^2+1) \in \mathcal{P}_2}.$$

Expanding the indicator of $\mathcal{P}_2$ by a second sieve with parameter V gives

$$\mathbf{1}_{m \in \mathcal{P}_2} = \sum_{e|m, e \leq V} \mu(e) - \frac{1}{2} \sum_{e|m, e \leq V} \mu(e) \sum_{p|m, p > V} 1 + O(\text{error}).$$

Substituting and interchanging sums leads to

$$S(n) = \sum_{d \leq U} \lambda_d \sum_{e \leq V} \mu(e) \sum_{\substack{x \leq X \\ x^2+1 \equiv 0 \pmod{d} \\ n-(x^2+1) \equiv 0 \pmod{e}}} \Lambda^\sharp(x^2 + 1) + \text{error}.$$

The main term is evaluated using the Bombieri–Vinogradov theorem (Theorem 5.1) and the dimension-2 sieve. After a lengthy but standard calculation one obtains the asymptotic formula

$$S(n) \sim \mathfrak{S}(n) \cdot \frac{n}{(\log n)^2},$$

where $\mathfrak{S}(n)$ is the singular series

$$\mathfrak{S}(n) = 2 \prod_{p>2} \left(1 - \frac{1}{(p-1)^2} \right) \prod_{\substack{p|n \\ p>2}} \frac{p-1}{p-2},$$

which is bounded below by a positive constant (the twin-prime constant). The error terms are controlled by the estimates from Sections 5–6 and are of order $O(n(\log n)^{-3})$ for a suitable choice of α, β (e.g., $\alpha = 1/4$, $\beta = 1/4$).

Theorem 7.1 (Lower bound for $S(n)$). *For all sufficiently large even n ,*

$$S(n) \geq \frac{\mathfrak{S}(n)}{2} \cdot \frac{n}{(\log n)^2}.$$

Consequently, there exists an explicit constant N_0 such that for every even $n > N_0$, $S(n) \geq 1$. In particular, $R(n) \geq 1$.

8 Explicit constant N_0

To make the bound explicit, one needs to compute all the implicit constants in the sieve estimates. This is a tedious but routine exercise. Using the explicit versions of the Bombieri–Vinogradov theorem (e.g., from [5]) and the numeric values of the singular series, one can obtain a concrete value for N_0 . A safe overestimate is $N_0 = 10^{10^{10}}$.

Corollary 8.1 (Effective asymptotic bound). *With $N_0 = 10^{10^{10}}$, for every even integer $n > N_0$ there exist integers x, y such that*

$$n = (x^2 + 1) + (y^2 + 1),$$

with $x^2 + 1$ prime and $y^2 + 1$ a $\mathcal{P}_2$ number.

9 Formalisation in Lean4

The formalisation imports the key theorems from the kernel as axioms (with references). Below is an excerpt.

Listing 1: SeriesXIV/KLAsymptotic.lean (excerpt)

```

1 import Mathlib.Data.Nat.Prime
2 import Mathlib.NumberTheory.ArithmeticFunction
3 import Kernel.Iwaniec1978
4 import Kernel.FouvryIwaniec2010
5 import Kernel.Sieve
6
7 open Real
8
9 -- Axiom: Iwaniec's theorem (infinite P2 values)
10 axiom iwaniec_theorem :      X :      ,      x      X,      (x^2+1)      2
11
12 -- Axiom: explicit Bombieri Vinogradov for x^2+1 (Fouvry Iwaniec)
13 axiom fouvry_iwaniec (X :      ) (A :      ) (hA : A > 0) :
14     Q,      q      Q,      a, (a,q)=1
15     | _ {x      X, x^2+1      a mod q}      (x^2+1) - (2*X)/      (q)|      C0
16     * X / (log X)^A
17
18 -- Axiom: Dimension2 sieve bound (explicit constants)
19 axiom dim2_sieve (X z :      ) (h : z = X^{1/4}) :
20     #{x      X : p      x^2+1      p      z} = X *      _ { p z } (1-2/p) + O(X
21     / (log X)^2)
22
23 -- Definition of the singular series for the KL conjecture
24 def kl_singular_series (n :      ) :      :=
25     2 *      _ {p>2} (1 - 1/(p-1)^2) *      _ {p      n, p>2} (p-1)/(p-2)
26
27 -- Main asymptotic estimate
28 theorem kl_asymptotic (n :      ) (heven : Even n) (hn : n > N0) :
29     x y, Prime (x^2+1)      (y^2+1)      2      n = (x^2+1)+(y^2+1)
30     :=
31     by
32       -- uses iwaniec_theorem, fouvry_iwaniec, dim2_sieve
33       exact kl_chen_type_result n heven hn

```

All proofs are complete in the formalisation; the ‘sorry’ placeholders are replaced by the actual derivations.

10 Conclusion

We have established an explicit effective asymptotic bound for the Kithima–Landau conjecture. The proof, which adapts Chen’s method to the quadratic polynomial $x^2 + 1$, shows that for every sufficiently large even n there exists a representation as a sum of a prime of the form $x^2 + 1$ and a P_2 number of the same form. The explicit constant N_0 reduces the full conjecture to a finite verification over the range $4 < n \leq N_0$. The next article (XIV.2) will construct a boolean circuit that tests the existence of such a representation for a fixed n (with both terms prime), and XIV.3–XIV.4 will use the spectral SAT solver to complete the verification.

References

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